Below is a **master table of important AI terminologies**, **all major model types**, and their **real-time software use cases with examples**.

✅ Note: "List of *all* AI terms" is practically infinite (new terms keep coming), but this covers **almost every term used in real AI projects, interviews, papers, and production systems**.

✅ AI Terminology + Model Types + Real-Time Use Cases (Master Table)

Category	Terminology / Model Type	Meaning (Detailed)	Real-time Use Cases	Example in Software / Application
AI Basics	Artificial Intelligence (AI)	Systems that perform tasks needing human intelligence	Automation, reasoning, decision support	Chatbots, fraud detection systems
	Machine Learning (ML)	Learns patterns from data to predict/decide	Prediction, classification, recommendations	Netflix movie recommendation
	Deep Learning (DL)	ML using multi-layer neural networks	Images, NLP, speech	Face unlock, OCR, translation
	Data Science	Extract insights from data using stats/ML	Analytics, dashboards, forecasting	Sales forecasting dashboard
	Model	A trained function that maps input → output	All AI apps	Spam classifier model
	Algorithm	Method/process to learn patterns	Training models	Gradient descent training
	Dataset	Collection of training/test data	Training AI	Customer transaction records
	Feature	Input variables for ML model	Feature engineering	Age, salary used for loan approval
	Label	Output/class in supervised learning	Classification tasks	Spam/Not spam

	Prediction	Model output inference result	Business automation	Predict churn risk
	Inference	Using trained model in production	Real-time ML APIs	/predict endpoint
	Training	Learning parameters from data	Model building	Training CNN on images
	Retraining	Training again with new data	Drift handling	Retrain fraud model weekly
Learning Types	Supervised Learning	Learn with labeled data	Classification/regression	Email spam detection
	Unsupervised Learning	Learn hidden patterns without labels	Clustering, anomaly	Customer segmentation
	Semi-Supervised Learning	Few labeled + many unlabeled	Medical imaging	Disease detection with limited labels
	Self-Supervised Learning	Model creates labels itself	Foundation models	BERT, GPT training
	Reinforcement Learning (RL)	Agent learns by rewards/punishment	Robotics, games, optimization	Self-driving simulation training
Model Outputs	Classification	Predict category/class	Fraud / spam / diagnosis	"Fraud vs Not fraud"
	Regression	Predict continuous value	Forecasting, pricing	House price prediction
	Ranking	Sort results by relevance	Search engines	Google search ranking
	Clustering	Group similar items	Segmentation	Group customers by behavior
	Anomaly Detection	Identify unusual events	Fraud, cyber attacks	Credit card anomaly detector

	Recommendation	Suggest best items	E-commerce	Amazon product suggestions
Traditional ML Models	Linear Regression	Predict continuous value linearly	Forecasting	Predict sales based on ads
	Logistic Regression	Classification using sigmoid	Binary classification	Loan approval
	Decision Tree	Rule-based split model	Interpretable ML	Medical decision support
	Random Forest	Many trees ensemble	Robust classification	Fraud detection
	Gradient Boosting	Sequential tree improvement	High accuracy models	XGBoost credit scoring
	XGBoost / LightGBM	Optimized boosting algorithms	Production ML widely used	Bank risk scoring
	SVM (Support Vector Machine)	Optimal margin classifier	Small/medium datasets	Handwriting recognition
	Naive Bayes	Probabilistic classifier	Text classification	Spam filters
	KNN	Classification by nearest neighbors	Similarity search	Recommend similar products
	K-Means	Clustering algorithm	Customer clustering	Segment users into 5 groups
Neural Network Models	ANN (Neural Network)	Basic neural net	Classification	Image category classifier
	DNN	Deep neural network	Complex patterns	Voice detection
	CNN	Vision network for images	Image/video tasks	Face detection, OCR
	RNN	Sequence model	Time series, text	Stock prediction (simple)

	LSTM / GRU	Improved RNN	Long sequences	Speech-to-text
	Transformer	Attention-based model	NLP + multimodal	ChatGPT, BERT
NLP / LLM Models	Bag of Words	Count-based text representation	Simple NLP	Keyword classifier
	TF-IDF	Word importance score	Search, classification	Document similarity
	Word2Vec / GloVe	Word embeddings	NLP tasks	Similar words search
	Embedding Model	Converts text/image to vectors	Search, RAG	Semantic search engine
	BERT	Encoder transformer	Classification, extraction	Sentiment analysis
	GPT	Decoder transformer	Generative text	Chatbots, content generation
	LLM (Large Language Model)	Very large transformer trained on internet-scale text	Chat, code, agents	ChatGPT in customer support
	Fine-Tuning	Train existing LLM on domain data	Domain-specific assistant	Medical/legal assistant
	Instruction Tuning	Teach LLM to follow instructions	Better chat assistants	“Answer politely” tuning
	RLHF	Reinforcement learning from human feedback	Align responses	Safer chatbot behavior
	Alignment	Make model safe & useful	Guardrails	Filter harmful responses
Generative AI	Generative Model	Generates new data/content	Text/image/video /music	AI image generator
	GAN	Generator vs discriminator	Image generation	Fashion product mockups

	VAE	Latent variable generative	Variations generation	Face morph generation
	Diffusion Model	Denoising-based generation	Image/video	Stable Diffusion-style models
	Prompt Engineering	Designing best input prompts	Improve results	"Act as doctor…" prompt
	Negative Prompt	Remove unwanted concepts	Image gen	"no blur, no watermark"
Computer Vision	Image Classification	Detect what object is	Sorting	Detect "cat/dog"
	Object Detection	Detect object + bbox	Surveillance, retail	YOLO in CCTV
	YOLO	Real-time object detector	Fast detection	Helmet detection on site
	Segmentation	Pixel-level classification	Medical, autonomous driving	Tumor segmentation
	OCR	Extract text from images	Documents	Aadhaar/PAN extraction
Audio / Speech AI	Speech-to-Text (ASR)	Convert voice to text	Call centers	Live transcription
	Text-to-Speech (TTS)	Convert text to voice	Assistants	AI voice support agent
	Speaker Identification	Recognize who is speaking	Security	Voice login
Recommenders	Collaborative Filtering	Recommend using user similarity	E-commerce	"Users like you bought…"
	Content-Based Filtering	Recommend similar items	Streaming	Suggest similar genre songs
	Hybrid Recommender	Mix both	Strongest approach	Netflix + Amazon use hybrid
Evaluation Metrics	Accuracy	Correct predictions ratio	Classification	Spam detection accuracy

	Precision	Correct positives / predicted positives	Fraud detection	Reduce false fraud alerts
	Recall	Correct positives / actual positives	Healthcare	Catch more disease cases
	F1 Score	Balance precision+recall	Imbalanced datasets	Fraud, anomaly detection
	ROC-AUC	Ranking performance	Classifier quality	Risk scoring
	Confusion Matrix	TP FP TN FN breakdown	Debugging model	Fraud model analysis
	MAE / MSE / RMSE	Regression error metrics	Forecast models	Demand prediction
	BLEU / ROUGE	NLP generation quality	Summarization	Auto summary evaluation
Production AI	MLOps	DevOps for ML models	Deployment lifecycle	CI/CD for model
	Pipeline	Automated workflow	Training + deploy	Data→train→deploy
	Feature Store	Central place for features	Reuse features	Same features for training + inference
	Model Registry	Stores versioned models	Governance	MLflow registry
	Monitoring	Track drift + performance	Reliability	Alert when accuracy drops
	Data Drift	Input data changes	Detect changes	New customer behavior patterns
	Concept Drift	Relationship changes (X→Y)	Model decay	Fraud strategies change
	A/B Testing	Compare two models	Optimization	Model v1 vs v2 conversion rate
	Shadow Deployment	Run new model silently	Safe rollout	Compare predictions quietly

	Canary Release	Release to small % first	Safe deployment	5% traffic to new model
	Latency	Response time	Real-time	Loan decision in 50ms
	Throughput	Requests per second	High-scale	10k RPS chatbot
	Scalability	Handle increasing load	Microservices	Autoscaling inference servers
	Edge AI	Model runs on device	Low latency	Mobile camera AI
AI APIs	REST API for inference	AI model exposed as API	Software integration	/predict endpoint
	Batch Inference	Process bulk input	Offline ML	Nightly churn predictions
	Streaming Inference	Real-time stream input	Live analytics	Kafka fraud detection
Explainability	Explainable AI (XAI)	Interpretable AI decisions	Compliance	Loan approval explanation
	SHAP	Feature contribution scores	Explain predictions	Why user got rejected
	LIME	Local explanation	Debugging	Explain single prediction
Security / Safety	Adversarial Attack	Manipulate input to fool AI	Cybersecurity	Alter image to bypass detector
	Prompt Injection	Trick LLM instructions	LLM security	Inject hidden malicious prompt
	Jailbreak	Bypass safety restrictions	LLM safety	Unsafe content generation attempts
	Guardrails	Safety filters	Compliance	Profanity + policy check
	PII	Personal identifiable info	Data compliance	Mask phone/email in logs

RAG Systems	RAG (Retrieval Augmented Generation)	LLM answers using retrieved documents	Enterprise assistants	"Company policy chatbot"
	Vector DB	Stores embeddings	Semantic search	Pinecone/FAISS/ Weaviate
	Chunking	Split docs into pieces	Better retrieval	Split PDF into 500-token chunks
	Similarity Search	Find closest embeddings	Search engine	Match query with docs
	Re-ranking	Improve retrieval results	Accuracy boost	Cross-encoder reranking
Agents	AI Agent	LLM that performs tasks	Automation	Booking tickets bot
	Tool Calling	LLM calls APIs/tools	Multi-step tasks	LLM calls Weather API
	Planning	Break tasks into steps	Long workflows	"Plan travel & book hotels"
	Memory	Store context	Personalization	Remember user preferences
	Autonomy	Executes tasks without constant user input	Productivity	AI operations assistant
Real AI Applications	Fraud Detection	Detect suspicious transactions	FinTech	Real-time card fraud
	Churn Prediction	Predict customer leaving risk	Telecom/SaaS	Retention campaign trigger
	Predictive Maintenance	Predict machine failure	Manufacturing	Sensor-based warning system
	Credit Scoring	Risk evaluation	Banking	Approve/reject loans
	Recommendation Engine	Suggest items	E-commerce/media	Product/movie/music suggestions
	Chatbot / Virtual Assistant	Automated support	Customer service	WhatsApp AI support agent

	Document AI	Understand PDFs/invoices	Finance/HR	Invoice extraction & validation
	Medical Imaging AI	Diagnose from scans	Healthcare	Cancer detection from X-ray
	Cybersecurity AI	Detect threats/anomalies	Security	Intrusion detection system

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✅ Summary of "All Model Types" (Quick Categories)

To make it easier:

1) Traditional ML

- Logistic Regression, Random Forest, XGBoost
  ✅ Best for: structured/tabular business data (banking, insurance)

2) Deep Learning

- CNN, RNN/LSTM/GRU, Transformers
  ✅ Best for: images, speech, NLP, large unstructured data

3) Generative AI

- GPT, Diffusion, GAN
  ✅ Best for: content generation, coding assistants, images/video/music

4) Reinforcement Learning

- Q-Learning, DQN, PPO
  ✅ Best for: sequential decision-making (robotics, games, supply chain)

5) Hybrid Production AI Systems

- RAG + Vector DB + LLM + Agents
  ✅ Best for: enterprise AI assistants, search, policy bots

Below is a **500+ AI terminology glossary (A–Z)** used in **AI / ML / DL / NLP / Generative AI / LLM / MLOps / RAG / Agents / CV / Speech / Security**.

 Format: **Term — Short meaning (real-world context)**
If you want, I can also provide this in **CSV / Excel** format for study + interview prep.

✅ 500+ AI Terminology Glossary (A–Z)

A (50+)

1. **Activation Function** — Function applied to neuron output (ReLU, Sigmoid, Tanh)
2. **AdaGrad** — Optimizer adapting learning rate per parameter
3. **Adam** — Popular adaptive optimizer (fast training, stable)
4. **AdamW** — Adam with correct weight decay handling
5. **Ablation Study** — Remove components to see impact
6. **Accuracy** — Correct predictions / total predictions
7. **Action Space** — All possible agent actions in RL
8. **Active Learning** — Model asks for labels for uncertain samples
9. **Actor-Critic** — RL method using actor policy + critic value estimator
10. **Adversarial Attack** — Trick model by small perturbations
11. **Adversarial Example** — Modified input that fools model
12. **Adversarial Training** — Train model using adversarial samples
13. **Agent** — AI system that observes + acts to achieve goals
14. **AGI** — Artificial General Intelligence (human-level across tasks)
15. **ALBERT** — Lightweight version of BERT
16. **Alignment** — Make model outputs safe/useful/human-preferred
17. **AlphaGo** — RL system beating human Go champions
18. **AlphaFold** — Protein structure prediction model
19. **Analytics** — Extract insights/statistics from data
20. **Anchor Boxes** — Predefined detection boxes in object detection
21. **Annealing** — Gradually reduce learning rate/temperature
22. **Anomaly Detection** — Detect unusual patterns/events
23. **Apache Spark MLlib** — ML library for big data
24. **API Latency** — Response time of inference API
25. **Approximate Nearest Neighbor (ANN)** — Fast similarity search technique
26. **ARIMA** — Classic time series forecasting model
27. **Artificial Neural Network (ANN)** — Basic neural network model
28. **ASR** — Automatic Speech Recognition (speech→text)
29. **Attention** — Mechanism to focus on relevant context
30. **Attention Head** — One parallel attention unit in transformer
31. **Attribute** — Feature/property of an entity
32. **AUC (Area Under Curve)** — Classification ranking performance metric
33. **Augmentation** — Create modified training samples (flip, noise)
34. **Autoencoder** — Neural net compressing then reconstructing data
35. **Autoregressive Model** — Predict next token based on previous
36. **AutoML** — Automated model selection + tuning
37. *A Search** — Path finding algorithm (also used in planning)
38. **ASGI** — Async server gateway interface (ML APIs)
39. **Audio Embedding** — Numeric vector representation of sound
40. **Annotation** — Labeling data for supervised training
41. **AI Governance** — Policies controlling model usage compliance
42. **AI Safety** — Prevent harmful AI behavior
43. **AI Ethics** — Fairness, transparency, privacy issues
44. **ArcFace** — Face recognition loss function

45. **Axial Attention** — Attention variant for high-dim inputs
46. **Adversarial Robustness** — Model stability against attacks
47. **Affective Computing** — AI understanding emotions
48. **Autonomous System** — System acts without human
49. **Agnostic Model** — Model not tied to a specific assumption
50. **Adaptation** — Modifying model to new domain
51. **Amplitude** — Audio wave magnitude
52. **Artificial Dataset** — Synthetic data

B (50+)

53. **Backpropagation** — Compute gradients for NN training
54. **Bagging** — Ensemble technique reducing variance
55. **Bag of Words (BoW)** — Text represented by word counts
56. **Batch** — Group of samples processed together
57. **Batch Normalization** — Normalize layer inputs for stability
58. **Beam Search** — Search strategy for best output sequences
59. **Bias (Model)** — Systematic error tendency
60. **Bias (NN)** — Intercept term in neurons
61. **Bias-Variance Tradeoff** — Underfit vs overfit balancing
62. **Big Data** — High volume/velocity/variety data
63. **Binary Classification** — Two-class prediction task
64. **Binarization** — Convert values to 0/1
65. **BiLSTM** — Bidirectional LSTM sequence model
66. **BERT** — Transformer encoder for NLP tasks
67. **BLEU Score** — Machine translation evaluation metric
68. **Booster** — Weak learner in boosting ensemble
69. **Boosting** — Sequential models improving residuals
70. **Bounding Box** — Rectangle around detected object
71. **BRNN** — Bidirectional recurrent neural network
72. **BPR (Bayesian Personalized Ranking)** — Ranking loss for recommender
73. **Bottleneck Layer** — Low-dimensional representation layer
74. **Bootstrap Sampling** — Random sampling with replacement
75. **Byte Pair Encoding (BPE)** — Tokenization technique
76. **Baseline Model** — Simple model for comparison
77. **Backbone Network** — Core feature extractor in CV
78. **Batch Inference** — Predict in bulk jobs offline
79. **Bayes Theorem** — Probability update rule
80. **Bayesian Optimization** — Hyperparameter tuning strategy
81. **Bayesian Network** — Probabilistic graphical model
82. **Behavior Cloning** — Imitation learning in RL
83. **Benchmark Dataset** — Standard dataset for comparing models
84. **Bidirectional Attention** — Context from both directions
85. **Bit Error Rate (BER)** — Communication signal metric (speech/IoT)
86. **BLEURT** — Learned NLP evaluation metric
87. **BLOOM** — Open multilingual LLM
88. **Bloom Filter** — Fast membership test, probabilistic

89. **BOW Encoder** — Encode text using BoW vector
90. **Branching Factor** — Tree search expansion count
91. **Brain-inspired Computing** — Neuromorphic AI
92. **Broadcasting (Tensor)** — Auto shape expansion in math ops
93. **Brittleness** — Model failing on slight changes
94. **Bucket (Data)** — Storage partition (features/logs)
95. **Buffer (RL)** — Replay memory for experiences
96. **Byte-level Tokenization** — Tokens are bytes not words
97. **Batch Size** — Training samples per gradient update
98. **Bilingual Model** — Supports 2 languages
99. **Binarized Neural Network** — Uses binary weights for efficiency
100. **Bias Mitigation** — Reduce unfairness in model decisions
101. **Backtesting** — Historical simulation for strategies
102. **Business Rules Engine** — Rule-based automation (with AI)

C (60+)

103. **Calibration** — Make predicted probabilities accurate
104. **Canary Deployment** — Deploy to small traffic portion
105. **Categorical Feature** — Feature with categories
106. **Categorical Cross Entropy** — Loss for multi-class classification
107. **Causal Inference** — Understand cause-effect not correlation
108. **Chain of Thought (CoT)** — Multi-step reasoning in LLMs
109. **Checkpoint** — Saved model weights
110. **Class Imbalance** — One class dominates dataset
111. **Class Weight** — Weight minority class more during training
112. **Classification** — Predict class/category
113. **Clipping** — Limit gradients to prevent explosion
114. **Closed-form Solution** — Exact formula solution (linear regression)
115. **CLIP** — Multimodal model mapping text+image embeddings
116. **Cloud Inference** — Model served in cloud
117. **Cluster** — Group of similar points
118. **Clustering** — Unsupervised grouping technique
119. **CNN** — Convolutional neural network for vision
120. **Cold Start** — Recommender problem for new users/items
121. **Collaborative Filtering** — Recommend based on similar users
122. **Convergence** — Training stabilizes at minimum
123. **Confusion Matrix** — TP FP TN FN evaluation table
124. **Context Window** — Tokens LLM can attend to
125. **Contrastive Learning** — Learn representations by comparing positives/negatives
126. **Control Variate** — Reduce variance in estimators
127. **Cosine Similarity** — Similarity measure for vectors
128. **Cost Function** — Another name for loss
129. **Covariate Shift** — Input distribution changes (drift)
130. **Cross Attention** — Attention between two sequences (encoder-decoder)

131. **Cross-Encoder** — Jointly encodes query+doc for ranking
132. **Cross Validation** — Evaluate with multiple train/test splits
133. **CUDA** — GPU compute platform used in deep learning
134. **Curriculum Learning** — Train from easy→hard samples
135. **Customer Churn** — Predict user leaving behavior
136. **CatBoost** — Gradient boosting library
137. **CTC Loss** — Speech recognition alignment loss
138. **CIFAR-10** — Standard image dataset
139. **COCO Dataset** — Object detection benchmark dataset
140. **Cohen's Kappa** — Agreement metric
141. **Contextual Bandit** — RL simplified setting (one-step decision)
142. **Continuous Learning** — Continual/online learning over time
143. **Concept Drift** — Relation between X and Y changes over time
144. **Compression** — Reduce model size
145. **Compute Graph** — Graph of operations in NN
146. **Conditional Generation** — Generate output given condition/prompt
147. **Conditional Random Field (CRF)** — NLP structured prediction model
148. **Catastrophic Forgetting** — Forget old tasks after learning new
149. **Chunking** — Splitting docs for RAG retrieval
150. **Chunk Overlap** — Repeated tokens between chunks
151. **Cache (KV Cache)** — Store attention keys/values for faster decoding
152. **Continual Learning** — Learning without retraining from scratch
153. **Content Moderation** — Filter harmful outputs
154. **Cost-Sensitive Learning** — Different penalties for errors
155. **Cramming** — Overfitting due to memorizing
156. **Convolution** — Sliding kernel operation in CNN
157. **Contrastive Loss** — Loss for similarity learning
158. **Cyclic LR** — Cyclic learning rate schedule
159. **Cumulative Reward** — Total reward in RL
160. **Computer Vision (CV)** — AI working on images/videos
161. **CORS** — Web security rule for APIs
162. **Counterfactual** — "What if?" alternate scenario analysis

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D (50+)

163. **Data Augmentation** — Expand data artificially
164. **Data Drift** — Input distribution change in production
165. **Data Leakage** — Train/test contamination causing fake performance
166. **Data Pipeline** — ETL flow feeding ML training/inference
167. **Data Labeling** — Human/auto process of annotation
168. **Data Normalization** — Scaling values (0–1, z-score)
169. **Data Preprocessing** — Cleaning & transformation steps
170. **Dataset Split** — Train/validation/test separation
171. **Decision Boundary** — Separation surface between classes
172. **Decision Tree** — Tree-based model
173. **Decoder** — Model that generates output sequence
174. **Deep Learning** — Neural nets with multiple layers

175. **Dense Layer** — Fully connected NN layer
176. **Deployment** — Put model into production
177. **Diffusion Model** — Generates by denoising process
178. **Dimensionality Reduction** — PCA/UMAP to reduce features
179. **Discriminator** — GAN network that judges real/fake
180. **Distillation** — Train smaller model using larger model outputs
181. **Distribution Shift** — Any change between train and test distributions
182. **DQN** — Deep Q-Network (RL)
183. **Dropout** — Randomly drop neurons (regularization)
184. **Dynamic Batching** — Combine requests for efficiency
185. **Domain Adaptation** — Move model to new domain
186. **Domain Shift** — Data changes across environments
187. **Decoder-only Transformer** — GPT-style LLM
188. **Dataloader** — Loads batches for training
189. **Data Warehouse** — Structured analytics store
190. **Data Lake** — Raw storage of all types data
191. **Dead Neuron** — ReLU neuron stuck at 0
192. **Dequantization** — Convert quantized weights to float
193. **Data Governance** — Rules for data security/quality
194. **Divergence (KL)** — Measure distribution difference
195. **DeBERTa** — Improved BERT variant
196. **Dense Retrieval** — Retrieval using embeddings
197. **Deterministic Policy** — Fixed action for a state
198. **Differentiable** — Supports gradient computation
199. **Discretization** — Convert continuous → categorical
200. **Distance Metric** — Euclidean, cosine, etc.
201. **Downsampling** — Reduce sample rate/data volume
202. **Dual Encoder** — Encode query/doc separately for retrieval

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E (45+)

203. **Early Stopping** — Stop training when val loss stops improving
204. **Embedding** — Vector representation of text/image/audio
205. **Embedding Space** — High-dimensional vector space
206. **ELMo** — Deep contextual word representation model
207. **Encoder** — Model processing input to latent representation
208. **Encoder-Decoder** — Seq2seq transformer architecture
209. **Ensemble** — Combine multiple models
210. **Epoch** — Full pass through dataset
211. **Entropy** — Measure of uncertainty
212. **Epsilon-Greedy** — RL exploration strategy
213. **ER (Experience Replay)** — Store experiences for training (RL)
214. **Evaluation** — Measure model performance
215. **Exploding Gradient** — Gradients become huge
216. **Explainability** — Understanding model decisions
217. **Explanations** — SHAP/LIME-based feature effects
218. **Edge AI** — On-device inference (mobile/IoT)

219. **EER (Equal Error Rate)** — Speaker verification metric
220. **Embedding Model** — Model generating embeddings (e.g., text embedding)
221. **Entity Recognition** — NER extract names/locations
222. **ETL** — Extract Transform Load data
223. **Eigenvector** — PCA/linear algebra concept
224. **Elastic Net** — Regression with L1+L2 regularization
225. **ELU** — Activation function
226. **Energy-based Model** — Assign energy score to data
227. **Episodic Memory** — Agent memory for episodes
228. **Episodic RL** — RL tasks with episodes
229. **Expected Value** — Statistical expectation
230. **Embedding Cache** — Store embeddings for faster RAG
231. **Enrichment** — Adding extra data fields/features
232. **Error Analysis** — Study failure cases
233. **Expert System** — Rule-based AI system
234. **Experiment Tracking** — Record runs/metrics (MLflow)
235. **Exploration** — RL trying new actions
236. **Exploitation** — RL using best-known action
237. **Exponential Smoothing** — Time series technique

F (45+)

238. **False Positive (FP)** — Incorrectly predicted positive
239. **False Negative (FN)** — Incorrectly predicted negative
240. **Feature Engineering** — Create meaningful input features
241. **Feature Store** — Central storage for features
242. **Feature Scaling** — Normalize/standardize features
243. **Federated Learning** — Train on devices without sharing raw data
244. **Fine-tuning** — Adapt pretrained model to task/domain
245. **F1 Score** — Harmonic mean of precision & recall
246. **Few-shot Learning** — Learn from few examples
247. **Flatten Layer** — Convert multi-dim to 1D
248. **Flops** — Floating-point operations count
249. **Foundation Model** — Large pretrained multi-task model
250. **Forward Pass** — Compute output from inputs
251. **Focal Loss** — Addresses class imbalance (detection)
252. **Forecasting** — Predict future values
253. **Free-form Generation** — Unconstrained output generation
254. **FP16 / BF16** — Half precision formats for speed
255. **Full Fine-tuning** — Train all parameters
256. **Function Calling** — LLM calling external tools/APIs
257. **Fusion** — Combine signals (text+image)
258. **Fairness** — Avoid bias against groups
259. **Face Recognition** — Identify person from face
260. **Feature Importance** — Contribution of feature
261. **FAISS** — Vector similarity search library
262. **FLOPs Budget** — Compute constraint

263. **Fact Checking** — Validate statements
264. **Failure Mode** — Common error pattern
265. **Flash Attention** — Efficient transformer attention algorithm

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G (45+)

266. **GAN** — Generative adversarial network
267. **Gated Activation** — Uses gating (GLU)
268. **Gaussian Noise** — Random noise injection
269. **Generalization** — Works well on unseen data
270. **Generative AI** — AI that creates new content
271. **Generative Model** — Model producing samples
272. **Generator (GAN)** — GAN net generating fake data
273. **Gradient Descent** — Optimization technique
274. **Gradient Boosting** — Boosting for trees
275. **Gradient Clipping** — Prevent exploding gradients
276. **Gradient Explosion** — Too-large gradients
277. **Gradient Vanishing** — Gradients become too small
278. **Graph Neural Network (GNN)** — Learning on graphs (nodes/edges)
279. **GPU** — Hardware for parallel computation
280. **GloVe** — Word embeddings technique
281. **Ground Truth** — True labels
282. **Grid Search** — Hyperparameter brute-force search
283. **Guardrails** — Safety control layer for LLM outputs
284. **GMM** — Gaussian mixture model
285. **Gini Impurity** — Decision tree split metric
286. **Gated Recurrent Unit (GRU)** — RNN variant
287. **Graph Embedding** — Vector representation of graph data
288. **Geo-spatial AI** — AI for location data
289. **Global Average Pooling** — CNN pooling layer
290. **Greedy Decoding** — Take max probability token

H (35+)

291. **Hallucination** — LLM generates incorrect confident output
292. **Hard Negative** — Difficult negative sample for contrastive learning
293. **Hashing Trick** — Reduce dimensionality for text features
294. **Heterogeneous Data** — Mixed data types
295. **Hidden Layer** — Internal NN layers
296. **Hidden State** — RNN memory state
297. **Hierarchical Clustering** — Tree-like clustering algorithm
298. **High Bias** — Underfitting
299. **High Variance** — Overfitting
300. **Hinge Loss** — Loss used in SVM
301. **HNSW** — Efficient ANN graph indexing for vector DB

302. **Human-in-the-loop** — Human review for quality/safety
303. **Hyperparameter** — Config not learned (lr, batch size)
304. **Hyperparameter Tuning** — Optimize config values
305. **Homogeneous Data** — Same-type data
306. **He Initialization** — Weight init for ReLU
307. **Hugging Face Transformers** — Popular NLP library
308. **Heuristic** — Rule of thumb method
309. **Hessian** — Second derivative matrix

I (40+)

310. **Imbalanced Dataset** — Unequal class distribution
311. **Image Classification** — Predict image class
312. **Image Segmentation** — Pixel-wise classification
313. **Inference** — Use model to predict in production
314. **Information Retrieval** — Search relevant documents
315. **Input Pipeline** — Data loading for model
316. **Instruction Tuning** — Tune LLM to follow instructions
317. **Interpretability** — Model transparency
318. **IoU (Intersection over Union)** — Object detection metric
319. **IR Metrics** — Precision@k, Recall@k, NDCG
320. **Isolation Forest** — Anomaly detection algorithm
321. **Imitation Learning** — Learn behavior from experts
322. **Incremental Learning** — Learn with new data gradually
323. **In-Context Learning** — LLM learning inside prompt
324. **Indexing (Vector)** — Build ANN index for fast retrieval
325. **Information Gain** — Decision tree splitting criterion
326. **Inference Server** — Model serving infrastructure
327. **Int8 Quantization** — Compress weights to 8-bit
328. **Interpolation** — Estimate missing values
329. **IoT AI** — AI running on edge sensors
330. **Intent Classification** — Identify user intent
331. **Initialization** — Set starting weights
332. **Inner Product Search** — Similarity search method
333. **IRL** — Inverse Reinforcement Learning
334. **Instance Segmentation** — Segment each object instance

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J (15+)

335. **Jaccard Similarity** — Intersection/union similarity
336. **Jacobian** — Matrix of first derivatives
337. **Jailbreak** — Bypass LLM safety guardrails
338. **Java ML Serving** — Serving models in Java APIs
339. **JIT Compilation** — Runtime optimization (TensorRT/JVM)
340. **JSONL** — JSON lines format used in datasets

341. **Joint Embedding** — Embedding text+image together
342. **Joule** — Energy measure in compute cost
343. **Jensen-Shannon Divergence** — Symmetric KL divergence
344. **Just-in-time Data Prep** — Prep features at inference time

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K (25+)

345. **K-Means** — Clustering algorithm
346. **KNN** — k-nearest neighbors classification
347. **Kernel Trick** — SVM nonlinear mapping
348. **Knowledge Distillation** — Teacher→student model compression
349. **Knowledge Graph (KG)** — Graph of entities/relations
350. **Keyword Extraction** — Extract important words
351. **KL Divergence** — Distribution difference measure
352. **K-Fold CV** — Cross validation split technique
353. **KV Cache** — Key-value caching in transformer decoding
354. **Keras** — Deep learning library (high-level API)
355. **KPI** — Key performance indicator
356. **K-Anonymity** — Privacy mechanism
357. **Knowledge Base** — Stored information repository

L (45+)

358. **Label** — Target output
359. **Label Noise** — Incorrect labels
360. **Label Smoothing** — Prevent overconfidence
361. **Lambda (Regularization)** — Reg strength factor
362. **Latency** — Time per prediction
363. **LayerNorm** — Normalize layer activations
364. **Learning Rate (LR)** — Step size in optimization
365. **Learning Rate Scheduler** — Change LR over time
366. **Leaky ReLU** — ReLU variant
367. **Likelihood** — Probability of data given parameters
368. **Linear Regression** — Continuous prediction model
369. **Link Function** — In GLMs mapping linear output
370. **LLM** — Large language model
371. **LoRA** — Low-rank adaptation fine-tuning method
372. **Loss Function** — Training objective measure
373. **Log Loss** — Classification loss
374. **Logit** — Unnormalized score before softmax
375. **Logistic Regression** — Binary classifier
376. **Long Context** — LLM with large window
377. **Long Short-Term Memory (LSTM)** — RNN variant
378. **Label Encoder** — Convert categories to numbers
379. **Latent Space** — Hidden representation space

380. **Latent Variable Model** — Uses latent variables (VAE)
381. **Local Minima** — Small minimum in loss landscape
382. **LIME** — Local model explanation tool
383. **Large-Scale Training** — Distributed training
384. **Language Model** — Predict next token probability
385. **LangChain** — Framework for LLM apps + agents
386. **LLaMA** — Meta's LLM family
387. **LightGBM** — Fast gradient boosting model

M (55+)

388. **Machine Learning** — Learning patterns from data
389. **Margin** — Separation gap in classification
390. **Masked Language Modeling (MLM)** — BERT training objective
391. **Max Pooling** — Reduce dimensions in CNN
392. **Mean Squared Error (MSE)** — Regression loss
393. **Mean Absolute Error (MAE)** — Regression metric
394. **Memory Network** — Models with memory component
395. **Metadata** — Data about data
396. **Metric Learning** — Learn distance/similarity
397. **Microservices** — Deploy AI as modular services
398. **MinMax Scaling** — Scale to [0,1]
399. **Mixture of Experts (MoE)** — Multi-expert neural architecture
400. **MLflow** — Experiment tracking + registry tool
401. **Model Drift** — Model performance degrades over time
402. **Model Registry** — Model version store
403. **Model Serving** — Deploy model in API
404. **Momentum** — Optimizer acceleration term
405. **Monte Carlo Methods** — Random sampling estimation
406. **Multiclass Classification** — Predict among multiple classes
407. **Multimodal AI** — Uses text + image + audio together
408. **Multilabel Classification** — Multiple labels per sample
409. **MLOps** — ML lifecycle operations
410. **Masked Attention** — Attention that hides future tokens
411. **Matrix Factorization** — Recommender method
412. **Multihead Attention** — Multiple attention heads
413. **Markov Decision Process (MDP)** — RL environment model
414. **Mean Reciprocal Rank (MRR)** — IR ranking metric
415. **Multitask Learning** — Learn multiple tasks jointly
416. **Model Compression** — Reduce model size
417. **Mixed Precision Training** — FP16/BF16 for speed
418. **Monotonicity Constraint** — ML constraint for interpretability
419. **Mamba** — State-space model alternative to transformers

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N (40+)

420. **Naive Bayes** — Probabilistic classifier
421. **Named Entity Recognition (NER)** — Extract names/locations
422. **NDCG** — Ranking metric
423. **Negative Sampling** — Efficient training method
424. **Neural Network** — Layered function approximator
425. **NeRF** — Neural radiance field (3D generation)
426. **Normalization** — Scale/standardize data
427. **Noise** — Random variation in data
428. **Nonlinearity** — Activation enabling complex functions
429. **Node** — Unit in graph networks
430. **NLP** — Natural language processing
431. **Nonstationary** — Data distribution changes
432. **N-gram** — Sequence of n tokens
433. **Neural ODE** — Differential equation based networks
434. **NaN** — Not-a-number in numeric computations
435. **Network Pruning** — Remove weak weights
436. **Novelty Detection** — Unknown class detection
437. **Neuro-symbolic AI** — Combine NN + logic rules
438. **No Free Lunch Theorem** — No single model best for all tasks

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O (35+)

439. **Object Detection** — Locate objects in image
440. **Objective Function** — Optimize this during training
441. **Offline RL** — RL from logged datasets
442. **On-device Inference** — AI running locally
443. **Online Learning** — Update model continuously
444. **One-hot Encoding** — Convert category into binary vector
445. **One-shot Learning** — Learn from single example
446. **Open Set Recognition** — Identify unknown classes
447. **Optimizer** — Method to update weights
448. **Overfitting** — Performs well on train, poorly on unseen
449. **OOD Detection** — Out-of-distribution detection
450. **Overparameterization** — Too many parameters
451. **OpenAI Function Calling** — Structured tool calls
452. **Orchestration** — Coordinating tasks/services
453. **Ontology** — Formal representation of knowledge

P (55+)

454. **Parameter** — Learnable weight in model
455. **PCA** — Dimensionality reduction
456. **Perceptron** — Basic linear classifier
457. **Perplexity** — LM quality metric (lower better)
458. **Pipeline** — Automated workflow

459. **Pooling** — Downsampling operation
460. **Precision** — TP / predicted positives
461. **Prompt** — Input text instruction for LLM
462. **Prompt Engineering** — Craft prompts to get best output
463. **Prompt Injection** — Attack manipulating LLM instructions
464. **Pretraining** — Train on large dataset first
465. **Post-training** — Alignment stage after pretraining
466. **Policy** — RL mapping state→action
467. **Policy Gradient** — RL using gradient updates to policy
468. **PPO** — Proximal Policy Optimization RL algorithm
469. **Precision@K** — Retrieval metric
470. **Probabilistic Model** — Outputs probability distributions
471. **Positional Encoding** — Adds token position info
472. **Pruning** — Remove parameters/neurons
473. **Productionization** — Make model ready for production
474. **Privacy** — Data protection requirement
475. **Pseudo-labeling** — Use model predictions as labels
476. **PyTorch** — Deep learning framework
477. **Pydantic** — Schema validation in Python APIs
478. **Propensity Score** — Causal inference score
479. **Parameter Efficient Fine-Tuning (PEFT)** — Fine-tune small subset (LoRA)
480. **Personalization** — Tailor model for user
481. **Prediction Interval** — Range of prediction confidence
482. **Parallelism (Data/Model)** — Multi-GPU distributed training
483. **Precision-Recall Curve** — PR tradeoff visualization
484. **Program Synthesis** — Generate code from intent

Q (20+)

485. **Q-Learning** — Value-based RL algorithm
486. **Q-Function** — Estimate value of actions
487. **Quantization** — Reduce weight precision for efficiency
488. **Query** — Search input
489. **Query Expansion** — Add related terms for search
490. **QKV** — Query-Key-Value attention components
491. **QLoRA** — Quantized LoRA fine-tuning
492. **Queue (Serving)** — Request queue in inference service
493. **Quorum** — Consensus requirement in distributed systems

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R (60+)

494. **RAG** — Retrieval Augmented Generation
495. **Random Forest** — Ensemble of decision trees
496. **Ranking** — Sorting results by relevance
497. **Recall** — TP / actual positives

498. **Recommender System** — Suggest items to users
499. **ReLU** — Common activation function
500. **Regression** — Predict numeric values
501. **Regularization** — Prevent overfitting (L1/L2/Dropout)
502. **Reinforcement Learning** — Learn by rewards
503. **Reward Function** — RL scoring mechanism
504. **Retrieval** — Search relevant chunks/documents
505. **Retriever** — Module performing retrieval in RAG
506. **Re-ranking** — Rank retrieved results again
507. **Robustness** — Stability under noise/changes
508. **RLHF** — Reinforcement learning from human feedback
509. **RNN** — Recurrent neural network
510. **RoBERTa** — Robust BERT variant
511. **ROC Curve** — Tradeoff curve (TPR vs FPR)
512. **ROUGE** — Summarization evaluation metric
513. **Root Cause Analysis** — Identify why model failed
514. **Runtime Optimization** — Speed up inference
515. **Red Teaming** — Testing model for failure attacks
516. **Rate Limiting** — Limit API request rate
517. **Ray** — Distributed compute framework
518. **Retraining Schedule** — Periodic training cycle
519. **Risk Scoring** — Predict risk probability
520. **Reward Shaping** — Modify reward to guide learning
521. **Residual Connection** — Skip connections in transformer
522. **Reproducibility** — Same results with same seeds/config
523. **Random Seed** — Starting point for randomness
524. **Regularized Loss** — Loss + penalty term
525. **Response Generation** — LLM output stage
526. **Reasoning Model** — LLM optimized for reasoning steps

S (60+)

527. **Sampling** — Choosing random outputs in generation
528. **Scaling Law** — Performance vs model/data/compute scaling
529. **Self-Attention** — Attention within same sequence
530. **Self-Supervised Learning** — Learn labels from data itself
531. **Semantic Search** — Search by meaning using embeddings
532. **Sentence Embedding** — Vector for sentence meaning
533. **Sequence-to-Sequence** — Map input sequence to output
534. **SGD** — Stochastic gradient descent optimizer
535. **SHAP** — Model explanation using Shapley values
536. **Siamese Network** — Twin networks for similarity learning
537. **Sigmoid** — Activation giving probabilities
538. **Similarity Search** — Find nearest vectors in embedding space
539. **SimCLR** — Contrastive representation learning method
540. **SMOTE** — Oversampling for imbalance
541. **Softmax** — Convert logits to probability distribution

542. **Sparse Attention** — Reduce attention computation
543. **Sparse Retrieval** — BM25 keyword retrieval
544. **Speech Recognition** — Convert voice → text
545. **Speech Synthesis** — Text → voice
546. **Stacking** — Ensemble using meta-model
547. **State Space** — All states in RL
548. **Stochastic Policy** — Probabilistic action selection
549. **Streaming Inference** — Real-time prediction on streams
550. **Stopwords** — Common words removed in NLP
551. **Supervised Learning** — Learn from labeled samples
552. **SVM** — Support vector machine
553. **Synthetic Data** — Artificial data generation
554. **Safety Filters** — Block unsafe content output
555. **Self-Consistency** — Multiple reasoning paths vote
556. **SFT (Supervised Fine-Tuning)** — Fine-tune on instruction datasets
557. **Speculative Decoding** — Faster generation method
558. **SentencePiece** — Tokenizer library
559. **Spacy** — NLP library
560. **Search Index** — Precomputed structure for retrieval
561. **Speech Embeddings** — Voice representation vectors
562. **SQL Feature Store** — Store features in DB
563. **System Prompt** — LLM hidden instruction context

T (55+)

564. **Tabular Data** — Structured data rows/columns
565. **Temperature** — Controls randomness in LLM output
566. **Tensor** — Multi-dimensional array
567. **TensorRT** — NVIDIA inference optimization
568. **Test Set** — Held-out data for evaluation
569. **Text Classification** — Sentiment/spam/intent
570. **TF-IDF** — Weighted word importance representation
571. **Token** — Small unit of text
572. **Tokenization** — Split text into tokens
573. **Top-K Sampling** — Sample among top K tokens
574. **Top-P (Nucleus) Sampling** — Sample among smallest set reaching prob p
575. **Transfer Learning** — Reuse pretrained model
576. **Transformer** — Attention-based deep learning architecture
577. **Training Loop** — Batch→loss→backprop step process
578. **Training Data** — Dataset for learning
579. **Trajectory** — Sequence of RL states/actions
580. **True Positive (TP)** — Correct predicted positive
581. **True Negative (TN)** — Correct predicted negative
582. **TTS** — Text-to-speech
583. **Time Series** — Time-based data sequence
584. **Teacher Forcing** — Feed true previous token during training
585. **Threshold** — Decision boundary cutoff

586. **Token Budget** — Max tokens allowed
587. **Tool Calling** — LLM calling external tools
588. **Training Drift** — Data issues in training process
589. **Token Embedding** — Vector mapping for tokens
590. **Trajectory Optimization** — RL planning method
591. **Toxicity** — Harmful content detection
592. **Triton Inference Server** — NVIDIA model serving

U (25+)

593. **U-Net** — Segmentation CNN architecture
594. **Underfitting** — Model too simple
595. **Unsupervised Learning** — Learn from unlabeled data
596. **User Profiling** — Build user preference model
597. **Uncertainty Estimation** — Measure prediction confidence
598. **UPR (Unsupervised PR)** — PR without labels
599. **UMAP** — Dimensionality reduction method
600. **Unit Test (ML)** — Validate pipeline/model code

V (35+)

601. **Validation Set** — Data used for tuning
602. **Variance** — Sensitivity to training data
603. **Vector** — Numeric list representing object
604. **Vector Database** — Store + search vectors
605. **Vector Index** — ANN structure for vector search
606. **VAE** — Variational autoencoder
607. **Vanishing Gradient** — Gradients go to 0
608. **Vision Transformer (ViT)** — Transformer for images
609. **Vocab** — All tokens set
610. **Vocabulary Size** — Token count in tokenizer
611. **Voice Bot** — TTS+ASR AI agent
612. **Value Function** — Expected reward from state
613. **ViLT** — Vision-and-language transformer model
614. **Vector Quantization** — Discrete latent embedding method
615. **Variance Reduction** — Reduce estimator noise
616. **Viterbi Algorithm** — Decode best sequence in HMM/CRF
617. **Volatility Forecasting** — Financial time-series forecasting

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W (20+)

618. **WandB (Weights & Biases)** — Experiment tracking
619. **Warm Start** — Continue training from checkpoint

620. **Warmup** — Gradually increase LR early training
621. **Weight Decay** — Regularization penalty on weights
622. **Word Embedding** — Vector representation of words
623. **Word2Vec** — Word embedding training method
624. **WER (Word Error Rate)** — Speech recognition metric
625. **Weak Supervision** — Noisy labels from heuristics
626. **Whisper** — Speech recognition model
627. **Windowing** — Segment time-series or audio in windows
628. **World Model** — RL predictive environment model

X (10+)

629. **XAI** — Explainable AI
630. **XGBoost** — Gradient boosting library
631. **XML Annotation** — Dataset label format
632. **XLM-R** — Multilingual transformer model
633. **X-Entropy** — Cross entropy loss
634. **XOR Problem** — Classic non-linear NN example

Y (10+)

635. **YOLO** — Real-time object detection model
636. **YAML Config** — Configuration format used in ML projects
637. **Yield (RL)** — Performance metric in RL tasks

Z (10+)

638. **Z-Score** — Standardization measure
639. **Zero-shot Learning** — Solve task without task-specific training
640. **ZSL (Zero Shot Learning)** — Another term for zero-shot
641. **Z-Transform** — Signal processing technique
642. **Zigzag Decoding** — Variant decoding strategy in seq models

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Below is the **complete 642 AI terms glossary in TABLE format (directly here)**.

✅ 642 AI Terms Glossary (Table)

#	Term	Meaning (short + practical)

1	Activation Function	Function applied to neuron output (ReLU/Sigmoid/Tanh).
2	AdaGrad	Optimizer adapting learning rate per parameter.
3	Adam	Adaptive optimizer widely used in deep learning.
4	AdamW	Adam variant with correct weight decay.
5	Ablation Study	Removing components to check impact on performance.
6	Accuracy	Correct predictions / total predictions.
7	Action Space	Set of all possible actions in RL.
8	Active Learning	Model asks for labels on uncertain samples.
9	Actor-Critic	RL method combining policy (actor) + value model (critic).
10	Adversarial Attack	Input manipulation to fool model.
11	Adversarial Example	Perturbed input that breaks model prediction.
12	Adversarial Training	Train using adversarial samples for robustness.
13	Agent	System that observes, decides, and acts.
14	AGI	Artificial General Intelligence (human-like general ability).
15	ALBERT	Lightweight BERT variant.
16	Alignment	Making model outputs safe, helpful, human-preferred.
17	AlphaGo	Deep RL system mastering the game Go.
18	AlphaFold	Protein structure prediction model.
19	Analytics	Extract insights from data patterns and trends.
20	Anchor Boxes	Predefined bounding boxes used in detection.
21	Annealing	Gradual reduction of learning rate/temperature.
22	Anomaly Detection	Detect unusual behavior or rare events.
23	Apache Spark MLlib	Distributed ML library for big data pipelines.
24	API Latency	Response time of inference endpoint.
25	Approximate Nearest Neighbor (ANN)	Fast similarity search for embeddings.
26	ARIMA	Statistical time-series forecasting model.
27	Artificial Neural Network (ANN)	Multi-layer neuron model for prediction.
28	ASR	Automatic Speech Recognition (speech → text).
29	Attention	Mechanism focusing on relevant input parts.
30	Attention Head	One attention unit inside multi-head attention.
31	Attribute	Property/feature of an entity.
32	AUC	Area Under ROC Curve metric for classification ranking.
33	Augmentation	Create new training samples (flip/crop/noise).
34	Autoencoder	NN learning compressed latent representation.
35	Autoregressive Model	Predict next token/value from previous context.
36	AutoML	Automates model selection, tuning, pipelines.
37	A* Search	Pathfinding/planning algorithm.
38	ASGI	Async Python interface for ML APIs (FastAPI).
39	Audio Embedding	Vector representation of audio/speech.
40	Annotation	Human labeling process for supervised learning.
41	AI Governance	Policies for safe/legal AI deployment.

42	AI Safety	Prevent harmful or unintended AI behavior.
43	AI Ethics	Fairness, bias, transparency, accountability.
44	ArcFace	Loss method improving face recognition separation.
45	Axial Attention	Efficient attention for high-dimensional inputs.
46	Adversarial Robustness	Model resistance to adversarial perturbations.
47	Affective Computing	AI that detects/interprets emotions.
48	Autonomous System	System acting without constant human input.
49	Agnostic Model	Model not dependent on assumptions about data.
50	Adaptation	Adjusting model for a new dataset/domain.
51	Amplitude	Audio wave magnitude (signal strength).
52	Artificial Dataset	Synthetic dataset generated for training/testing.

#	Term	Meaning
53	Backpropagation	Compute gradients for NN learning.
54	Bagging	Ensemble technique via bootstrap aggregating.
55	Bag of Words (BoW)	Represent text using word counts.
56	Batch	Group of samples processed in one step.
57	Batch Normalization	Normalize activations to stabilize training.
58	Beam Search	Decoding strategy to explore top sequences.
59	Bias (Model)	Systematic model error due to assumptions/data.
60	Bias (NN)	Learnable intercept parameter in neuron.
61	Bias-Variance Tradeoff	Balance underfitting vs overfitting.
62	Big Data	Data large in volume, speed, variety.
63	Binary Classification	Prediction among 2 classes.
64	Binarization	Converting values to 0/1.
65	BiLSTM	LSTM reading sequences forward and backward.
66	BERT	Transformer encoder pretrained for NLP.
67	BLEU Score	Metric for translation quality evaluation.
68	Booster	Weak learner in boosting model.
69	Boosting	Sequential learning improving errors/residuals.
70	Bounding Box	Rectangle around object in detection.
71	BRNN	Bidirectional recurrent neural network.
72	BPR	Ranking loss for recommender systems.
73	Bottleneck Layer	Compression layer in encoder architectures.
74	Bootstrap Sampling	Sample dataset with replacement.
75	Byte Pair Encoding (BPE)	Tokenization using byte merge rules.
76	Baseline Model	Simple comparison model.
77	Backbone Network	Feature extractor in CNN/detection models.
78	Batch Inference	Predict in bulk offline mode.
79	Bayes Theorem	Conditional probability update rule.
80	Bayesian Optimization	Smart hyperparameter tuning method.
81	Bayesian Network	Graph model of probabilistic relations.
82	Behavior Cloning	Learn policy by imitating expert behavior.
83	Benchmark Dataset	Standard dataset for comparison.
84	Bidirectional Attention	Attention from both directions.

85	Bit Error Rate (BER)	Signal decoding error metric.
86	BLEURT	Learned metric for NLP generation evaluation.
87	BLOOM	Open multilingual large language model.
88	Bloom Filter	Probabilistic structure for membership test.
89	BOW Encoder	Text encoding using bag-of-words.
90	Branching Factor	Expansion rate in search tree.
91	Brain-inspired Computing	Neuromorphic computing approaches.
92	Broadcasting (Tensor)	Auto-expanding tensor shapes in ops.
93	Brittleness	Model failure on small perturbation.
94	Bucket (Data)	Storage partition in pipelines.
95	Buffer (RL)	Replay memory holding experiences.
96	Byte-level Tokenization	Tokenization at byte granularity.
97	Batch Size	Samples per training iteration.
98	Bilingual Model	Model trained to support 2 languages.
99	Binarized Neural Network	NN with binary weights for efficiency.
100	Bias Mitigation	Techniques reducing unfair bias.
101	Backtesting	Evaluate strategy/model on historical data.
102	Business Rules Engine	Rule-based automation often combined with AI.

#	Term	Meaning (short + practical)
103	Calibration	Adjust probabilities so confidence matches real accuracy.
104	Canary Deployment	Release new model to small % users first.
105	Categorical Feature	Feature having categories (city, gender, plan type).
106	Categorical Cross Entropy	Loss used for multi-class classification.
107	Causal Inference	Identify cause-effect (not just correlation).
108	Chain of Thought (CoT)	Step-by-step reasoning style for LLM output.
109	Checkpoint	Saved model weights during/after training.
110	Class Imbalance	Dataset dominated by one class (fraud vs normal).
111	Class Weight	Give different weights to classes in loss calculation.
112	Classification	Predict category/label (spam/not spam).
113	Clipping	Limiting gradients/values to stabilize training.
114	Closed-form Solution	Exact math solution without iterative training.
115	CLIP	Multimodal model aligning image-text embeddings.
116	Cloud Inference	Model predictions served on cloud infrastructure.
117	Cluster	Group of similar data points.
118	Clustering	Unsupervised grouping of data.
119	CNN	Convolutional neural network (image processing).
120	Cold Start	No data for new user/item in recommender systems.
121	Collaborative Filtering	Recommending via user-user or item-item similarity.
122	Convergence	Training reaches stable minimum of loss.
123	Confusion Matrix	Evaluation table (TP/FP/TN/FN).
124	Context Window	Maximum tokens an LLM can use at once.
125	Contrastive Learning	Learn representations using positive vs negative pairs.
126	Control Variate	Reduce estimator variance (in RL/probability).

127	Cosine Similarity	Vector similarity metric used in embeddings.
128	Cost Function	Same as loss function (optimization objective).
129	Covariate Shift	Input distribution changes across environments.
130	Cross Attention	Attention between different sequences (enc-dec).
131	Cross-Encoder	Model jointly encodes query+doc for reranking.
132	Cross Validation	Multiple train/test splits for robust evaluation.
133	CUDA	GPU computing platform (NVIDIA).
134	Curriculum Learning	Train from easy examples → harder ones.
135	Customer Churn	Customer leaves/cancels subscription.
136	CatBoost	Gradient boosting library optimized for categorical features.
137	CTC Loss	Loss for alignment in speech recognition.
138	CIFAR-10	Benchmark dataset with 10 image classes.
139	COCO Dataset	Large benchmark dataset for detection/segmentation.
140	Cohen's Kappa	Agreement metric beyond chance.
141	Contextual Bandit	RL where each decision is one step (no long episode).
142	Continuous Learning	Model learns continuously over time.
143	Concept Drift	Relationship between input and output changes.
144	Compression	Reduce model size (quantization/pruning).
145	Compute Graph	Graph of operations for forward/backprop.
146	Conditional Generation	Generate output conditioned on input/prompt.
147	Conditional Random Field (CRF)	Structured prediction for sequences (NER tags).
148	Catastrophic Forgetting	Model forgets old tasks when learning new ones.
149	Chunking	Split docs into small blocks for RAG retrieval.
150	Chunk Overlap	Shared tokens between chunks to keep context.
151	Cache (KV Cache)	Store transformer keys/values for faster decoding.
152	Continual Learning	Learn new tasks without full retraining.
153	Content Moderation	Filter toxic / illegal / unsafe content.
154	Cost-Sensitive Learning	Errors have different costs (fraud FN > FP).
155	Cramming	Memorization effect causing overfitting.
156	Convolution	Sliding filter operation in CNN.
157	Contrastive Loss	Loss function used for similarity learning.
158	Cyclic LR	Learning rate varies cyclically over epochs.
159	Cumulative Reward	Total reward over time/episode in RL.
160	Computer Vision (CV)	AI for image/video understanding.
161	CORS	Web restriction rule affecting API access.
162	Counterfactual	"What if scenario?" alternative outcome analysis.

#	Term	Meaning
163	Data Augmentation	Create new training samples from existing data.
164	Data Drift	Input data changes after deployment.
165	Data Leakage	Training accidentally uses future/test info.
166	Data Pipeline	Flow: collect → process → train/infer.
167	Data Labeling	Assign correct outputs (ground truth).
168	Data Normalization	Scaling numeric values consistently.

169	Data Preprocessing	Clean, format, transform raw data.
170	Dataset Split	Divide into train/val/test.
171	Decision Boundary	Line/surface separating classes.
172	Decision Tree	Tree model using rules/splits.
173	Decoder	Model part generating output sequence.
174	Deep Learning	ML using deep neural networks.
175	Dense Layer	Fully connected neural network layer.
176	Deployment	Putting model into production.
177	Diffusion Model	Generates data by denoising noise step-by-step.
178	Dimensionality Reduction	Reduce features (PCA/UMAP).
179	Discriminator	GAN network that detects real vs fake.
180	Distillation	Train small model using big model outputs.
181	Distribution Shift	Train vs production data mismatch.
182	DQN	Deep Q-Network RL algorithm.
183	Dropout	Randomly disable neurons for regularization.
184	Dynamic Batching	Group requests dynamically to improve throughput.
185	Domain Adaptation	Adapt model to a different domain dataset.
186	Domain Shift	Different environments produce different data.
187	Decoder-only Transformer	GPT-style transformer architecture.
188	Dataloader	Component that loads data batches efficiently.
189	Data Warehouse	Structured storage for analytics.
190	Data Lake	Storage for raw structured/unstructured data.
191	Dead Neuron	ReLU neuron stuck outputting 0 always.
192	Dequantization	Convert quantized numbers back to float-like form.
193	Data Governance	Rules/policies for data access/quality/security.
194	Divergence (KL)	Measure difference between distributions.
195	DeBERTa	Transformer model improving BERT.
196	Dense Retrieval	Embedding-based document retrieval.
197	Deterministic Policy	Same action always for same state.
198	Differentiable	Supports gradient computation for learning.
199	Discretization	Convert continuous values into buckets/categories.
200	Distance Metric	Method to compute similarity/distance.
201	Downsampling	Reduce sample rate or dataset size.
202	Dual Encoder	Encodes query and doc separately for retrieval.

#	Term	Meaning
203	Early Stopping	Stop training when validation stops improving.
204	Embedding	Vector representation (text/image/audio).
205	Embedding Space	Where embeddings lie, similarity meaningful.
206	ELMo	Contextual word embeddings (pre-transformer).
207	Encoder	Converts input into internal representation.
208	Encoder-Decoder	Transformer seq2seq style (T5).
209	Ensemble	Combine predictions of multiple models.
210	Epoch	One full pass through training dataset.
211	Entropy	Uncertainty/impurity measure.
212	Epsilon-Greedy	RL explore with probability ε else exploit.

213	Experience Replay (ER)	Store and reuse past RL experiences.
214	Evaluation	Measure model performance on unseen data.
215	Exploding Gradient	Very large gradients destabilizing training.
216	Explainability	Ability to explain model decisions.
217	Explanations	Human-understandable reason for predictions.
218	Edge AI	Run inference on device (mobile/IoT).
219	EER	Equal Error Rate in verification systems.
220	Embedding Model	Model producing embeddings (e.g., sentence embedding).
221	Entity Recognition	Extract entities like names/places.
222	ETL	Extract → Transform → Load data.
223	Eigenvector	Principal direction in linear algebra/PCA.
224	Elastic Net	Regression with both L1 and L2 penalties.
225	ELU	Activation function alternative to ReLU.
226	Energy-based Model	Learns energy scores for configurations.
227	Episodic Memory	Store past episodes for learning/agents.
228	Episodic RL	RL tasks with defined episode endings.
229	Expected Value	Average expected outcome.
230	Embedding Cache	Store precomputed embeddings for speed.
231	Enrichment	Adding extra features/data fields.
232	Error Analysis	Analyze failure patterns/edge cases.
233	Expert System	Rule-based AI (if-else reasoning).
234	Experiment Tracking	Log runs, metrics, params (MLflow/W&B).
235	Exploration	Trying new actions/behaviors in RL.
236	Exploitation	Using best-known action in RL.
237	Exponential Smoothing	Time-series smoothing for forecasting.

#	Term	Meaning
238	False Positive (FP)	Model predicts positive incorrectly.
239	False Negative (FN)	Model misses a true positive case.
240	Feature Engineering	Create meaningful input variables.
241	Feature Store	Central repo of reusable features.
242	Feature Scaling	Normalize/standardize numeric features.
243	Federated Learning	Train across devices without sharing raw data.
244	Fine-tuning	Adapt pretrained model for a task/domain.
245	F1 Score	Balance of precision and recall.
246	Few-shot Learning	Learn with very few labeled examples.
247	Flatten Layer	Convert multi-dim tensor to 1D vector.
248	FLOPs	Approx compute cost in operations.
249	Foundation Model	Large pretrained multi-purpose model.
250	Forward Pass	Run inputs through model to compute output.
251	Focal Loss	Loss to handle class imbalance in detection.
252	Forecasting	Predict future values from time-series.
253	Free-form Generation	Unstructured output generation.
254	FP16 / BF16	Half precision numeric formats for speed.
255	Full Fine-tuning	Update all weights during fine-tuning.

256	Function Calling	LLM calls external tools/APIs.
257	Fusion	Combine signals (text+image+audio).
258	Fairness	Avoid discriminatory outcomes.
259	Face Recognition	Identify person from face image.
260	Feature Importance	Contribution of feature to prediction.
261	FAISS	Similarity search library for embeddings.
262	FLOPs Budget	Compute constraint for model training/inference.
263	Fact Checking	Verify generated statements.
264	Failure Mode	Common model error behavior.
265	Flash Attention	Fast memory-efficient attention algorithm.

#	Term	Meaning
266	GAN	Generative adversarial network.
267	Gated Activation	Activation controlled by gates (GLU).
268	Gaussian Noise	Random noise added to improve robustness.
269	Generalization	Performance on unseen data.
270	Generative AI	AI creating new content (text/image/video).
271	Generative Model	Model producing new samples.
272	Generator (GAN)	GAN network producing fake data.
273	Gradient Descent	Iterative optimization method.
274	Gradient Boosting	Sequential tree ensemble technique.
275	Gradient Clipping	Limit gradient size for stability.
276	Gradient Explosion	Training instability due to huge gradients.
277	Gradient Vanishing	Gradients too small to learn.
278	Graph Neural Network (GNN)	Neural nets on graph-structured data.
279	GPU	Hardware for parallel computation.
280	GloVe	Word embeddings method based on co-occurrence.
281	Ground Truth	True correct labels.
282	Grid Search	Brute-force hyperparameter tuning.
283	Guardrails	Safety layer restricting LLM behavior.
284	GMM	Gaussian mixture model clustering.
285	Gini Impurity	Decision tree split impurity measure.
286	GRU	RNN variant with gating for memory.
287	Graph Embedding	Vector encoding for graph nodes/edges.
288	Geo-spatial AI	AI on geographic/location data.
289	Global Average Pooling	CNN pooling reducing spatial dims globally.
290	Greedy Decoding	Always select highest probability next token.

#	Term	Meaning
291	Hallucination	Confident but incorrect LLM output.
292	Hard Negative	Challenging negative sample for training.
293	Hashing Trick	Map features into fixed-size space efficiently.
294	Heterogeneous Data	Data containing multiple types/modalities.
295	Hidden Layer	Internal neural network layer.

296	Hidden State	RNN memory representation.
297	Hierarchical Clustering	Cluster using tree/dendrogram approach.
298	High Bias	Underfitting behavior.
299	High Variance	Overfitting behavior.
300	Hinge Loss	Loss used in SVM margin classification.
301	HNSW	Graph-based ANN index for vector search.
302	Human-in-the-loop	Human review/feedback improves AI system.
303	Hyperparameter	Config value not learned (lr, batch size).
304	Hyperparameter Tuning	Search best hyperparameter combination.
305	Homogeneous Data	Single-type consistent data.
306	He Initialization	Weight init optimized for ReLU networks.
307	Hugging Face Transformers	Popular transformer model library.
308	Heuristic	Rule-based shortcut strategy.
309	Hessian	Second-derivative matrix in optimization.

#	Term	Meaning (short + practical)
310	Imbalanced Dataset	Dataset where one class dominates (fraud rare).
311	Image Classification	Predict class label of an image.
312	Image Segmentation	Pixel-wise labeling of an image.
313	Inference	Using trained model to predict in production.
314	Information Retrieval	Search relevant docs/records for a query.
315	Input Pipeline	Steps that load/transform data for training.
316	Instruction Tuning	Fine-tune LLM to follow instructions better.
317	Interpretability	Ability to understand model behavior/decisions.
318	IoU	Intersection over Union metric for detection overlap.
319	IR Metrics	Metrics for retrieval/search effectiveness.
320	Isolation Forest	Tree-based anomaly detection algorithm.
321	Imitation Learning	Learn behavior by copying expert demonstrations.
322	Incremental Learning	Train gradually on new data batches.
323	In-Context Learning	LLM learns pattern from prompt examples.
324	Indexing (Vector)	Building vector index (HNSW/IVF) for search.
325	Information Gain	Decision tree split score based on entropy.
326	Inference Server	Infrastructure that serves ML models (API).
327	Int8 Quantization	Compress weights/activations to 8-bit.
328	Interpolation	Estimate missing values between known points.
329	IoT AI	AI running on sensors/edge devices.
330	Intent Classification	Detect user intent in NLP chatbot systems.
331	Initialization	Setting initial NN weights before training.
332	Inner Product Search	Similarity search using dot product.
333	IRL	Inverse Reinforcement Learning (infer rewards).
334	Instance Segmentation	Detect + segment each object separately.

#	Term	Meaning
335	Jaccard Similarity	Similarity = intersection/union of sets.

336	Jacobian	Matrix of partial derivatives (gradients).
337	Jailbreak	Prompt technique to bypass safety filters.
338	Java ML Serving	Deploying models via Java Spring APIs.
339	JIT Compilation	Runtime optimization improving execution speed.
340	JSONL	JSON Lines dataset format (1 JSON per line).
341	Joint Embedding	Shared embedding space for multiple modalities.
342	Joule	Energy unit used in compute/power metrics.
343	Jensen-Shannon Divergence	Symmetric KL-divergence for comparing distributions.
344	Just-in-time Data Prep	Compute features only when inference is needed.

#	Term	Meaning
345	K-Means	Clustering algorithm partitioning into K clusters.
346	KNN	K-nearest neighbors classifier/regressor.
347	Kernel Trick	SVM method to solve nonlinear boundaries.
348	Knowledge Distillation	Train small student model from big teacher.
349	Knowledge Graph (KG)	Graph of entities and relations.
350	Keyword Extraction	Identify important keywords in text.
351	KL Divergence	Measure difference between two probability distributions.
352	K-Fold CV	Cross validation using K splits.
353	KV Cache	Cache attention keys/values for fast decoding.
354	Keras	High-level DL framework (TensorFlow API).
355	KPI	Business measure for success (conversion, churn).
356	K-Anonymity	Privacy method hiding identities in data.
357	Knowledge Base	Repository of structured domain knowledge.

#	Term	Meaning
358	Label	Correct target output (class/value).
359	Label Noise	Wrong/uncertain labels in dataset.
360	Label Smoothing	Reduce overconfidence in classification.
361	Lambda (Regularization)	Strength parameter controlling penalty.
362	Latency	Response time per request.
363	LayerNorm	Normalization used in transformers.
364	Learning Rate (LR)	Step size in training updates.
365	Learning Rate Scheduler	Strategy to change LR over training.
366	Leaky ReLU	ReLU allowing small negative slope.
367	Likelihood	Probability of observed data under a model.
368	Linear Regression	Predict continuous numeric output linearly.
369	Link Function	GLM mapping between linear output and response.
370	LLM	Large Language Model (GPT-type).
371	LoRA	Efficient fine-tuning using low-rank updates.
372	Loss Function	Objective minimized during training.
373	Log Loss	Loss for probabilistic classification models.

374	Logit	Raw score before applying sigmoid/softmax.
375	Logistic Regression	Classic classifier outputting probability.
376	Long Context	LLM with large token window (long docs).
377	LSTM	RNN variant handling long sequences.
378	Label Encoder	Convert categories into numeric IDs.
379	Latent Space	Hidden representation capturing patterns.
380	Latent Variable Model	Model assuming hidden variables (VAE).
381	Local Minima	Small basin in loss landscape.
382	LIME	Tool to explain individual predictions locally.
383	Large-Scale Training	Distributed training on many GPUs/TPUs.
384	Language Model	Predict probability of next word/token.
385	LangChain	Framework for LLM applications/agents/tools.
386	LLaMA	Meta's family of open LLMs.
387	LightGBM	Efficient gradient boosting decision trees.

#	Term	Meaning
388	Machine Learning	Learning patterns from data for prediction.
389	Margin	Distance between decision boundary and samples.
390	Masked Language Modeling (MLM)	BERT training task predicting masked words.
391	Max Pooling	CNN operation selecting max in region.
392	Mean Squared Error (MSE)	Regression loss (square error average).
393	Mean Absolute Error (MAE)	Regression metric using absolute errors.
394	Memory Network	NN architecture with explicit memory component.
395	Metadata	Data describing other data (schema, time, source).
396	Metric Learning	Learn embedding distance for similarity.
397	Microservices	Deploy AI in small independent services.
398	MinMax Scaling	Normalize values between 0 and 1.
399	Mixture of Experts (MoE)	Experts networks activated selectively.
400	MLflow	Tool for tracking experiments and models.
401	Model Drift	Model accuracy decreases over time.
402	Model Registry	Central place to version models.
403	Model Serving	Hosting model for inference (API).
404	Momentum	Optimizer term accelerating updates.
405	Monte Carlo Methods	Use random sampling to estimate values.
406	Multiclass Classification	Predict among many classes.
407	Multimodal AI	Model using multiple data types (text+image).
408	Multilabel Classification	Predict multiple labels for one input.
409	MLOps	ML lifecycle: train, deploy, monitor.
410	Masked Attention	Prevent attention to future tokens.
411	Matrix Factorization	Recommender technique decomposing rating matrix.
412	Multihead Attention	Parallel attention heads for richer features.
413	MDP	Markov Decision Process (RL environment model).
414	Mean Reciprocal Rank (MRR)	Ranking metric based on first correct result.
415	Multitask Learning	One model learns multiple tasks together.
416	Model Compression	Reduce size via pruning/quantization/distill.

417	Mixed Precision Training	Use FP16/BF16 for speed + memory saving.
418	Monotonicity Constraint	Force certain feature relationships in prediction.
419	Mamba	State-space model alternative to transformers.

#	Term	Meaning
420	Naive Bayes	Probability-based classifier (text spam).
421	Named Entity Recognition (NER)	Extract names, orgs, locations from text.
422	NDCG	Ranking metric emphasizing top results.
423	Negative Sampling	Train efficiently using negative examples.
424	Neural Network	Model using layers of neurons.
425	NeRF	Neural Radiance Fields for 3D reconstruction.
426	Normalization	Scale features for stable training.
427	Noise	Random variation; can harm or regularize model.
428	Nonlinearity	Allows network to learn complex mapping.
429	Node	Graph element representing entity/state.
430	NLP	Natural language processing tasks.
431	Nonstationary	Data distribution changes over time.
432	N-gram	Token sequence of length N.
433	Neural ODE	Neural nets defined via differential equations.
434	NaN	Not-a-number value in computation.
435	Network Pruning	Remove unnecessary weights/neurons.
436	Novelty Detection	Detect new/unknown patterns.
437	Neuro-symbolic AI	Combine neural nets with symbolic logic.
438	No Free Lunch Theorem	No model best for every problem.

#	Term	Meaning
439	Object Detection	Find objects and locations in images.
440	Objective Function	Function minimized/maximized in training.
441	Offline RL	Learn policy from stored logs/datasets.
442	On-device Inference	Run inference on mobile/edge device.
443	Online Learning	Update model continuously with new data.
444	One-hot Encoding	Encode category as binary vector.
445	One-shot Learning	Learn from just one labeled example.
446	Open Set Recognition	Handle unseen classes as “unknown”.
447	Optimizer	Algorithm updating weights to minimize loss.
448	Overfitting	Model memorizes train, fails on test.
449	OOD Detection	Detect when input is out-of-distribution.
450	Overparameterization	Too many params relative to data.
451	OpenAI Function Calling	Structured calling of external APIs/tools.
452	Orchestration	Coordinating services/workflows (agents/pipelines).
453	Ontology	Formal representation of knowledge concepts/relations.

#	Term	Meaning
454	Parameter	Learnable weight in model.
455	PCA	Reduce dimensions while keeping variance.
456	Perceptron	Simple linear classifier.
457	Perplexity	Language model quality metric (lower better).
458	Pipeline	Automated ML workflow sequence.
459	Pooling	Downsampling operation in CNNs.
460	Precision	TP / (TP + FP).
461	Prompt	Input instruction for LLM.
462	Prompt Engineering	Designing prompts to improve output.
463	Prompt Injection	Attack overriding system instructions.
464	Pretraining	Training on huge general dataset first.
465	Post-training	Training after pretraining (alignment).
466	Policy	RL strategy mapping state → action.
467	Policy Gradient	RL algorithm optimizing policy with gradients.
468	PPO	Popular stable RL algorithm.
469	Precision@K	Retrieval metric: correct results in top K.
470	Probabilistic Model	Outputs probability distributions.
471	Positional Encoding	Add token position info in transformer.
472	Pruning	Remove weights/neurons to compress model.
473	Productionization	Full process making model production-ready.
474	Privacy	Protect sensitive information (PII).
475	Pseudo-labeling	Use model predictions as training labels.
476	PyTorch	Deep learning framework.
477	Pydantic	Validation library for Python APIs.
478	Propensity Score	Probability of treatment assignment (causal).
479	PEFT	Fine-tune using minimal extra parameters.
480	Personalization	Tailor outputs based on user preferences.
481	Prediction Interval	Confidence range around predicted value.
482	Parallelism (Data/Model)	Distributed training strategy.
483	Precision-Recall Curve	Precision vs recall plot for classification.
484	Program Synthesis	Generate code/program from description.

#	Term	Meaning (short + practical)
485	Q-Learning	Value-based RL learning using Q-table/Q-values.
486	Q-Function	Function estimating value of state-action pair.
487	Quantization	Reduce precision (FP32→INT8) for speed/size.
488	Query	User input for retrieval/search system.
489	Query Expansion	Add related terms to improve retrieval results.
490	QKV	Query-Key-Value matrices used in attention.
491	QLoRA	LoRA fine-tuning on quantized models (efficient).
492	Queue (Serving)	Request queue buffering inference traffic.
493	Quorum	Minimum agreement needed in distributed consensus systems.

#	Term	Meaning
494	RAG	Retrieval-Augmented Generation: LLM + search docs.
495	Random Forest	Ensemble of decision trees for robust ML.
496	Ranking	Sorting results by relevance score.
497	Recall	TP / (TP + FN).
498	Recommender System	Suggest items/products/content to users.
499	ReLU	Activation function max(0,x).
500	Regression	Predict continuous numeric values.
501	Regularization	Reduce overfitting (L1/L2/Dropout).
502	Reinforcement Learning	Learn actions via reward feedback.
503	Reward Function	Defines success score for agent actions.
504	Retrieval	Fetch relevant documents/chunks for a query.
505	Retriever	Component retrieving relevant chunks for RAG.
506	Re-ranking	Second-stage ranking improving retrieval accuracy.
507	Robustness	Model stability against noise/adversarial input.
508	RLHF	Align LLM using human feedback + RL.
509	RNN	Recurrent neural network for sequences/time-series.
510	RoBERTa	Robustly optimized BERT variant.
511	ROC Curve	Plot of TPR vs FPR for classifiers.
512	ROUGE	Metric for summarization/translation evaluation.
513	Root Cause Analysis	Find real reason behind model failures.
514	Runtime Optimization	Improve inference speed (TensorRT, caching).
515	Red Teaming	Adversarial testing of model safety & failures.
516	Rate Limiting	Restrict max API calls per time window.
517	Ray	Distributed compute framework (AI pipelines).
518	Retraining Schedule	Plan for periodic model retraining.
519	Risk Scoring	Assign probability score to risk (fraud/loan).
520	Reward Shaping	Modify reward to guide RL learning faster.
521	Residual Connection	Skip connection improving training stability.
522	Reproducibility	Ability to reproduce results with same setup.
523	Random Seed	Seed controlling reproducible randomness.
524	Regularized Loss	Loss with penalty term (L1/L2).
525	Response Generation	Stage of LLM output creation.
526	Reasoning Model	Model optimized for step-by-step reasoning.

#	Term	Meaning
527	Sampling	Pick output tokens probabilistically (not greedy).
528	Scaling Law	Performance grows with data/model/compute scale.
529	Self-Attention	Tokens attend to each other in same sequence.
530	Self-Supervised Learning	Learn from unlabeled data using proxy tasks.
531	Semantic Search	Search by meaning using embeddings.
532	Sentence Embedding	Vector representing sentence meaning.
533	Sequence-to-Sequence	Model maps input seq → output seq (translation).
534	SGD	Stochastic Gradient Descent optimizer.
535	SHAP	Explain predictions with feature contributions.

536	Siamese Network	Twin network for similarity/verification tasks.
537	Sigmoid	Activation turning logits into 0–1 probability.
538	Similarity Search	Search nearest vectors (RAG / recommender).
539	SimCLR	Self-supervised contrastive learning framework.
540	SMOTE	Oversample minority class for imbalance handling.
541	Softmax	Convert logits to probability distribution.
542	Sparse Attention	Reduce attention compute using sparse patterns.
543	Sparse Retrieval	Keyword-based retrieval (BM25).
544	Speech Recognition	Convert audio to text (ASR).
545	Speech Synthesis	Convert text to natural speech (TTS).
546	Stacking	Ensemble method using meta-model outputs.
547	State Space	All possible environment states in RL.
548	Stochastic Policy	Action selection probabilistic (not fixed).
549	Streaming Inference	Predict on real-time streams (Kafka).
550	Stopwords	Common words removed in NLP preprocessing.
551	Supervised Learning	Learn using labeled data.
552	SVM	Support Vector Machine classifier.
553	Synthetic Data	Artificially generated training data.
554	Safety Filters	Block toxic/unsafe/illegal output content.
555	Self-Consistency	Generate multiple reasoning paths + vote.
556	SFT	Supervised Fine-Tuning of LLMs.
557	Speculative Decoding	Faster generation using draft model + verify.
558	SentencePiece	Tokenizer library for BPE/unigram models.
559	spaCy	NLP library for tokenization/NER/etc.
560	Search Index	Structured retrieval index for fast search.
561	Speech Embeddings	Vector representation of voice identity.
562	SQL Feature Store	Feature store built on SQL tables.
563	System Prompt	Hidden instructions controlling LLM behavior.

#	Term	Meaning
564	Tabular Data	Structured rows/columns business data.
565	Temperature	Controls randomness of LLM generation.
566	Tensor	Multi-dimensional array for deep learning.
567	TensorRT	NVIDIA inference acceleration toolkit.
568	Test Set	Held-out data for final evaluation.
569	Text Classification	Assign label to text (sentiment/spam/intent).
570	TF-IDF	Word importance representation for NLP.
571	Token	Small text unit used in LLM processing.
572	Tokenization	Split text into tokens.
573	Top-K Sampling	Choose next token from top K probable tokens.
574	Top-P Sampling	Choose from smallest token set reaching probability P.
575	Transfer Learning	Reuse pretrained model for new task.
576	Transformer	Attention-based NN architecture.
577	Training Loop	Repeat batch→loss→update process.
578	Training Data	Dataset used to learn patterns.

579	Trajectory	Sequence of RL states/actions.
580	True Positive (TP)	Correct predicted positive.
581	True Negative (TN)	Correct predicted negative.
582	TTS	Text-to-speech generation.
583	Time Series	Time-ordered data.
584	Teacher Forcing	Use true previous token during training.
585	Threshold	Cutoff value for classification decision.
586	Token Budget	Max input+output tokens allowed.
587	Tool Calling	LLM invoking external functions/tools.
588	Training Drift	Change in training data quality/meaning over time.
589	Token Embedding	Vector lookup for token ID.
590	Trajectory Optimization	Optimize paths in RL/robotics planning.
591	Toxicity	Harmful language measure.
592	Triton Inference Server	NVIDIA server for scalable model serving.

#	Term	Meaning
593	U-Net	CNN architecture for image segmentation.
594	Underfitting	Model too simple, fails even on training data.
595	Unsupervised Learning	Learn patterns without labels.
596	User Profiling	Model user preferences/behavior profile.
597	Uncertainty Estimation	Measure confidence of predictions.
598	UPR	Unsupervised Precision/Recall metrics.
599	UMAP	Dimensionality reduction + visualization technique.
600	Unit Test (ML)	Tests ensuring ML code correctness.

#	Term	Meaning
601	Validation Set	Dataset for tuning hyperparameters.
602	Variance	Sensitivity to training data changes.
603	Vector	Numeric representation list.
604	Vector Database	Store and search embeddings.
605	Vector Index	ANN data structure for fast vector search.
606	VAE	Variational Autoencoder generative model.
607	Vanishing Gradient	Gradients become too small to learn.
608	ViT	Vision Transformer for image tasks.
609	Vocab	All tokens in tokenizer dictionary.
610	Vocabulary Size	Number of tokens in vocabulary.
611	Voice Bot	Speech AI: ASR + LLM + TTS system.
612	Value Function	Expected reward estimate from state.
613	ViLT	Vision-language transformer model.
614	Vector Quantization	Compress embeddings into discrete codes.
615	Variance Reduction	Reduce estimation noise in learning.
616	Viterbi Algorithm	Best path decoding for HMM/CRF.
617	Volatility Forecasting	Predict market volatility using time-series.

#	Term	Meaning
618	W&B	Weights & Biases experiment tracking tool.
619	Warm Start	Continue training from earlier checkpoint.
620	Warmup	Gradually increase LR early in training.
621	Weight Decay	L2-style regularization term.
622	Word Embedding	Vector representation of words.
623	Word2Vec	Method to learn word embeddings.
624	WER	Word Error Rate for speech recognition accuracy.
625	Weak Supervision	Noisy labels from weak heuristics/rules.
626	Whisper	Speech recognition model (OpenAI).
627	Windowing	Split audio/time-series into windows.
628	World Model	Agent learns internal model of environment.

#	Term	Meaning
629	XAI	Explainable AI methods.
630	XGBoost	High-performance gradient boosting library.
631	XML Annotation	XML format for labeled datasets.
632	XLM-R	Multilingual transformer model.
633	X-Entropy	Cross entropy loss.
634	XOR Problem	Classic non-linear classification example.

#	Term	Meaning
635	YOLO	Real-time object detection model family.
636	YAML Config	Config format used for ML experiments.
637	Yield (RL)	Performance/output measure in RL setting.

#	Term	Meaning
638	Z-Score	Standardize value using mean/std deviation.
639	Zero-shot Learning	Solve task with no explicit training examples.
640	ZSL	Short form for Zero-shot Learning.
641	Z-Transform	Transform used in signal processing/time-series.
642	Zigzag Decoding	Variant decoding strategy in sequence models.